THE COLLISION FLUCTUATION SUM

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Abstract. For a prime $p > b+1$ not dividing the base b, the *collision fluctuation* at lag ℓ is $\Delta_p(\ell) = C(b^\ell \bmod p) - \overline{C}_p$, the deviation of the collision count from its mean over constructive multipliers. The partial sum

$$\Phi_x(s,\ell) = \sum_{\substack{b+1<p\le x \\ p\nmid b}} \frac{\Delta_p(\ell)}{p^s}$$

converges absolutely for $\mathrm{Re}(s) > 2$. Computation to 10^7 suggests Mertens-rate divergence at $s = 1$: $\Phi_x(1,\ell) \sim -\mu_\ell \cdot \log\log x$, where μ_ℓ depends on the lag and base ($\mu_1 \approx 0.68$ in base 10).

The divergence is controlled by the decomposition $\Delta_p = S - QR/(p-1-b)$, where $S = C - Q$ is an integer and $QR/(p-1-b)$ is a deterministic correction from the spectral class $R = (p-1) \bmod b$. The bilateral symmetry of the repetend forces $C(g)$ to be even for all g and constrains S to have the same parity as Q.

1. Introduction

This paper studies the fluctuation of the collision count across primes.

The preceding paper [3] decomposed the fractional field Dirichlet series into a mean part (controlled by the mean formula from the gate width paper [2]) and a fluctuation part. The mean part converges for $\mathrm{Re}(s) > 2$. The present paper studies the fluctuation part.

The classical prime number theorem is equivalent to the statement that $\sum \mu(n)/n$ converges, where μ is the Möbius function [5, 6]. This convergence encodes the cancellation of multiplicative structure across the integers at the edge of the critical strip.

Computation to 10^7 suggests that the collision fluctuation sum diverges at $s = 1$, growing like $\log\log x$ in a Mertens-type growth law [4]. The divergence arises from a persistent negative bias in the collision deviation: the specific multiplier b^ℓ is systematically below the mean collision count. A later paper investigates whether a centered version of the fluctuation sum converges at $s = 1$.

Date: October 2024.

2020 *Mathematics Subject Classification.* 11A63, 11N05, 11M41.

2. The Fluctuation Sum

Definition 1. For a prime $p > b+1$ not dividing b, the *collision fluctuation* at lag ℓ is

$$\Delta_p(\ell) = C(b^\ell \bmod p) - \overline{C}_p,$$

where $\overline{C}_p$ is the mean collision count over constructive nonidentity multipliers (those $g \neq 1$ with $C(g) > 0$). The *fluctuation partial sum* is

$$\Phi_x(s,\ell) = \sum_{\substack{b+1<p\le x \\ p\nmid b}} \frac{\Delta_p(\ell)}{p^s}.$$

The digit-partitioning prime $p = b+1$ (when prime) is excluded because $\overline{C}_p$ is undefined there ($p-1-b=0$).

Proposition 2 (Analytic convergence)**.** *The series* $\Phi(s,\ell) = \lim_{x\to\infty}\Phi_x(s,\ell)$ *converges absolutely for* $\mathrm{Re}(s) > 2$.

Proof. The fluctuation satisfies $|\Delta_p(\ell)| \le p-1$ trivially. Thus $|\Delta_p(\ell)|/p^s = O(p^{1-s})$. The sum $\sum_p p^{1-s}$ converges for $\mathrm{Re}(s) > 2$. □

Remark 3. The variance scaling [3] gives $\mathrm{Var}(\Delta) = O(Q)$ across multipliers at fixed p, suggesting typical $|\Delta_p| = O(\sqrt{Q}) = O(\sqrt{p/b})$. If this bound held pointwise for the specific multiplier b^ℓ, convergence would extend to $\mathrm{Re}(s) > 3/2$. However, the variance scaling is a statement about the distribution across multipliers, not a pointwise bound for a fixed lag. For a specific lag ℓ, the magnitude $|\Delta_p(\ell)|$ could be as large as Q.

3. The Exact Decomposition

Theorem 4 (Collision decomposition)**.** *The collision deviation satisfies*

$$\Delta_p = S - \frac{QR}{p-1-b},$$

where $S = C(b^\ell \bmod p) - Q$ *is an integer,* $Q = \lfloor (p-1)/b \rfloor$*, and* $R = (p-1) \bmod b$.

Proof. The mean collision count is $\overline{C}_p = Q(b(Q-1)+2R)/(b(Q-1)+R)$. Since $b(Q-1)+R = p-1-b$: $\overline{C}_p = Q + QR/(p-1-b)$. Therefore $\Delta_p = C - \overline{C}_p = (C-Q) - QR/(p-1-b) = S - QR/(p-1-b)$. □

Remark 5. Computation in base 10 at lag 1 shows that S takes values in $\{-9,-7,-4,-3,-2,-1,0,1,2,3,6,8\}$ for primes up to 10^7, with $S \in \{-1,0\}$ for approximately 60% of primes. Whether S is uniformly bounded as $p \to \infty$ is not established; the trivial bound is $|S| \le Q$.

Theorem 6 (Bilateral parity)**.** *For any prime $p > b$ and any $g \in \mathbb{F}_p^\times$, the collision count $C(g)$ is even.*

Proof. If r contributes to $C(g)$ (i.e., $\delta(r) = \delta(gr)$), then $p - r$ also contributes: $\delta(p-r) = b-1-\delta(r)$ and $\delta(g(p-r)) = \delta(-gr) = b-1-\delta(gr)$, so $\delta(p-r) = \delta(g(p-r))$. Since $r \neq p - r$ for $r \neq p/2$ (impossible for prime $p > 2$), the contributions pair. □

Corollary 7. *$S = C - Q$ has the same parity as Q.*

4. The Mertens Growth Law

Computation to 10^7 (664,574 primes) indicates that the partial sum $\Phi_x(1,1)$ diverges at $s = 1$, growing like $\log\log x$:

primes	$p_{\max}$	$\Phi_x(1,1)$	$\Phi_x/\log\log p$
1,000	7,951	-1.37	-0.623
10,000	104,779	-1.60	-0.652
100,000	1,299,811	-1.77	-0.671
664,574	9,999,991	-1.90	-0.682

Table 1. $\Phi_x(1,1)/\log\log p$ drifts steadily toward a constant $\mu_1 \approx 0.68$.

At smaller scales (up to 8,000 primes), the growth is invisible because $\log\log x$ grows extremely slowly: $\log\log(10^4) \approx 2.25$ versus $\log\log(10^7) \approx 2.78$.

Conjecture 8 (Mertens growth)**.** *The partial sum satisfies*

$$\Phi_x(1,\ell) = -\mu_\ell \cdot \log\log x + O(1)$$

for a constant $\mu_\ell > 0$ depending on the lag and base.

Observation 9. The transition from convergence to divergence is visible in the partial sums:

The divergence arises because Δ_p has a persistent negative mean: the mean of Δ_p over primes $p \leq x$ is approximately -0.91 in base 10 at $\ell = 1$, with 70% of primes having $\Delta_p < 0$, 20% having $\Delta_p > 0$, and 10% having $\Delta_p = 0$. The bilateral symmetry constrains $C(g)$ to be even but does not force Δ_p to have zero mean.

primes	$s = 1.5$	$s = 1.2$	$s = 1.0$	$s = 0.9$	$s = 0.5$
100	-0.17	-0.47	-1.07	-1.42	-6.97
500	-0.18	-0.55	-1.28	-2.02	-17.0
1000	-0.18	-0.56	-1.37	-2.17	-21.4
2000	-0.18	-0.57	-1.45	-2.30	-26.0

TABLE 2. $\Phi_x(s, 1)$ for various s. For $s > 1$, partial sums appear to stabilize. At $s = 1$, logarithmic growth. For $s < 1$, power-law growth.

p mod 10	R	mean S	mean $\overline{\Delta}$
1	0	-1.70	-1.70
3	2	-0.91	-1.11
7	6	-0.08	-0.68
9	8	$+0.68$	-0.12

TABLE 3. Per-class means for all primes up to 10^7 in base 10 at $\ell = 1$. All classes have negative mean $\overline{\Delta}$.

5. The Per-Class Structure

The collision deviation Δ_p decomposes by spectral class $a = p \bmod b$.

Observation 10. For each coprime class a, the mean of Δ_p over primes $p \equiv a \pmod b$ with $p \leq 10^7$ stabilizes to a class-specific value.

The per-class collision bias for the spectral class $R = 6$ (digit 7, the negative pole) is

$$\overline{\Delta}_{R=6} \approx -\frac{2}{3}.$$

The value 2/3 is the alignment limit of the prime 3 from the first paper [1]. The same constant appears as the collision bias of the negative polarity class.

The per-class mean of S in the complementary class $R = 8$ (digit 9, the neutral pole) is $\overline{S}_{R=8} \approx +2/3$. The bilateral symmetry exchanges these roles: the positive S in class 9 is absorbed by the spectral class correction $R/b = 0.8$, leaving a small negative $\overline{\Delta}$.

6. The Mertens Constant

The following heuristic relates the overall Mertens constant to the per-class biases.

By Mertens' theorem [4] for arithmetic progressions [5], each coprime class contributes equally to the prime sum at leading order. The Mertens constant is therefore approximately

$$\mu_\ell \approx \frac{1}{\varphi(b)} \sum_{\substack{a \bmod b \\ \gcd(a,b)=1}} |\overline{\Delta}_a|,$$

where φ is Euler's totient function.

In base 10 at $\ell = 1$: the average of the four class biases (1.70, 1.11, 0.68, 0.12) is 0.90. The observed ratio $\Phi_x(1,1)/\log\log x \approx 0.68$ is smaller; the discrepancy suggests that the absolute-value average overstates the effective constant, since the per-class contributions partially cancel in the signed sum. A precise derivation of μ_ℓ from the class means remains open.

The constant 2/3 from [1] appears not as the overall Mertens constant but as the per-class collision bias for the negative polarity class ($R = 6$, digit 7). The Mertens constant is a weighted average in which 2/3 is one of four terms.

7. Structure of the Cancellation

The partial cancellation in $\Phi_x(1,\ell)$ arises from the sawtooth structure of the collision count established in the preceding paper [3].

At each prime p, the collision count decomposes as $C(b^\ell \bmod p) = Q + S$, where S is a signed sum of Q terms, each ± 1. The sign of each term is determined by a comparison of two sawtooth sequences modulo p: one controlled by the collision parameter $c = b(1-b^\ell)^{-1} \bmod p$, the other by the base b.

As p ranges over the primes, the collision parameter c traces a path through the residues modulo p. The equidistribution of this path, controlled by the Möbius deranging family, determines how many of the Q signed terms are positive versus negative. The persistent negative bias means that c systematically occupies positions with more negative than positive sawtooth terms.

8. Remarks

The boundary at $s = 1$. The Riemann Hypothesis is equivalent to $\sum \mu(n)/n^s$ converging for all $\mathrm{Re}(s) > 1/2$. The collision fluctuation sum converges for $\mathrm{Re}(s) > 2$ and appears to diverge logarithmically at $s = 1$. Computation suggests that $s = 1$ is the boundary between convergence and divergence for $\Phi(s,\ell)$, with Mertens-type logarithmic growth rather than PNT-type convergence.

Forward direction. The divergence at $s = 1$ arises from a persistent per-class bias in Δ_p. The next paper investigates whether subtracting this bias (centering) produces a sum that converges at $s = 1$.

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